Evans, M. and Moshonov, H. (2006). "Checking for prior-data conflict." *Bayesian Analysis*, 1(4): 893–914. 2

Gelfand, A., Dey, D., and Chang, H. (1992). "Model determination using predictive distributions with implementation via sampling-based methods." In *Bayesian Statistics*. 7

Gelfand, A. E. and Ghosh, S. K. (1998). "Model choice: a minimum posterior predictive loss approach." *Biometrika*, 85(1): 1–11. 10

Gelman, A. (2004). "Exploratory data analysis for complex models." *Journal of Computational and Graphical Statistics*, 13(4): 755–779. 2, 6

Gelman, A., Meng, X.-L., and Stern, H. (1996). "Posterior predictive assessment of model fitness via realized discrepancies." *Statistica Sinica*, 733–760. 1, 2, 6, 10, 14, 15, 16, 18

Gelman, A. and Shalizi, C. (2012). "Philosophy and the practice of Bayesian statistics." *British Journal of Mathematical and Statistical Psychology*, 66: 8–38. 7

Gelman, A., Vehtari, A., Simpson, D., Margossian, C. C., Carpenter, B., Yao, Y., Kennedy, L., Gabry, J., Bürkner, P.-C., and Modrák, M. (2020). "Bayesian workflow." *arXiv preprint arXiv:2011.01808*. 2, 5, 9, 24

Gershman, S. and Goodman, N. (2014). "Amortized inference in probabilistic reasoning." *Proceedings of the Annual Meeting of the Cognitive Science Society*. 20

Guttman, I. (1967). "The use of the concept of a future observation in goodness-of-fit problems." *Journal of the Royal Statistical Society. Series B (Methodological)*, 83–100. 2, 6

Jeffreys, H. (1961). *The Theory of Probability*. Oxford University Press, 3 edition. 9

Jordan, M. I., Ghahramani, Z., Jaakkola, T. S., and Saul, L. K. (1999). "An introduction to variational methods for graphical models." *Machine Learning*, 37(2): 183–233. 19

Kass, R. E. and Raftery, A. E. (1995). "Bayes factors." *Journal of the American Statistical Association*, 90(430): 773–795. 9

Kingma, D. P. and Ba, J. (2014). "Adam: A method for stochastic optimization." *arXiv preprint arXiv:1412.6980*. 20

Kingma, D. P. and Welling, M. (2014). "Auto-encoding variational Bayes." *International Conference on Learning Representations*. 15, 18, 19

Marshall, E. and Spiegelhalter, D. (2007). "Identifying outliers in Bayesian hierarchical models: a simulation-based approach." *Bayesian Analysis*, 2(2): 409–444. 2, 7

Meng, X.-L. (1994). "Posterior predictive p-values." *The Annals of Statistics*, 22(3): 1142–1160. 1, 2

Newton, M. A. and Raftery, A. E. (1994). "Approximate Bayesian inference with the weighted likelihood bootstrap." *Journal of the Royal Statistical Society: Series B (Methodological)*, 56(1): 3–26. 27

O'Hagan, A. (2003). "HSSS model criticism (with discussion)." *Highly Structured Stochastic Systems*, 423–453. 2

Rezende, D. J., Mohamed, S., and Wierstra, D. (2014). "Stochastic backpropagation and approximate inference in deep generative models." In *International Conference on Machine Learning*, 1278–1286. PMLR. 19

Robins, J. M., van der Vaart, A., and Ventura, V. (2000). "Asymptotic distribution of p values in composite null models." *Journal of the American Statistical Association*, 95(452): 1143–1156. 2, 7

Rubin, D. B. (1984). "Bayesianly justifiable and relevant frequency calculations for the applied statistician." *The Annals of Statistics*, 12(4): 1151–1172. 1, 2, 6

Semenova, L., Rudin, C., and Parr, R. (2019). "A study in Rashomon curves and volumes: A new perspective on generalization and model simplicity in machine learning." *arXiv preprint arXiv:1908.01755*. 5

Stern, H. S., Arcus, D., Kagan, J., Rubin, D. B., and Snidman, N. (1995). "Using mixture models in temperament research." *International Journal of Behavioral Development*, 18(3): 407–423. 14, 15, 16

Tipping, M. E. and Bishop, C. M. (1999). "Probabilistic principal component analysis." *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 61(3): 611–622. 18

Vehtari, A. and Ojanen, J. (2012). "A survey of Bayesian predictive methods for model assessment, selection and comparison." *Statistics Surveys*, 6: 142–228. 10

Appendix A: Gaussian mixture model

Here, we provide details regarding the Gaussian mixture model in Section 1 and Section 2.2 of the main paper. The data $\mathbf{x}_i \in \mathbb{R}^D$, $i = 1, \ldots, N$ is generated from the following model with $D = 2$, $K^* = 3$ and $N = 500$.

$$\boldsymbol{\gamma}_i \sim \text{Multinomial}(\boldsymbol{p}, 1), \quad \boldsymbol{p} = \{1/K^*\}_{k=1}^{K^*} \tag{37}$$

$$\mathbf{x}_i | \boldsymbol{\gamma}_i, \boldsymbol{\mu}, \boldsymbol{\Sigma} \sim \sum_{k=1}^{K^*} \gamma_{ik} N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \text{where} \quad \boldsymbol{\Sigma}_k = \text{diag}\{\sigma_{k1}^2, \ldots, \sigma_{kd}^2\}, \tag{38}$$

with

$$\{\boldsymbol{\mu}_{dk}\}_{d,k=1}^{D,K^*} = \begin{pmatrix} -5 & 0 & 10 \\ 5 & 0 & 5 \end{pmatrix} \qquad \text{and} \qquad \{\sigma_{dk}^2\}_{d,k=1}^{D,K^*} = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & 4 \end{pmatrix}. \tag{39}$$

We fit the model:

$$\boldsymbol{\gamma}_i \sim \text{Multinomial}(\boldsymbol{p}, 1), \quad \boldsymbol{p} = \{1/K\}_{k=1}^{K} \tag{40}$$

$$\sigma_{kd}^2 \sim \text{Inverse-Gamma}(1, 1) \tag{41}$$

$$\boldsymbol{\mu}_k \sim N(0, 5^2) \tag{42}$$

$$\mathbf{x}_i | \boldsymbol{\gamma}_i, \boldsymbol{\mu}, \boldsymbol{\Sigma} \sim \sum_{k=1}^{K} \gamma_{ik} N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k). \tag{43}$$

Denote $\boldsymbol{\theta} = \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^{K}$. We set the true number of components to be $K^* = 3$. We run a PPN check to compare models with $K \in \{1, 2, 3, 4\}$ components. We fit each of these models using Gibbs sampling.

We conduct the partial predictive and PPN checks with the heldout diagnostic (12). The underlying realized diagnostic function is the log-likelihood, given by

$$d_K(\mathbf{x}, \boldsymbol{\theta}) = \sum_{k=1}^{K} \left\{ \sum_{i=1}^{n} \left\{ -\frac{1}{2} \gamma(\mathbf{x}_i, \boldsymbol{\theta})(\mathbf{x}_i - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k (\mathbf{x}_i - \boldsymbol{\mu}_k) \right\} - \frac{1}{2} \sum_{i=1}^{n} \gamma(\mathbf{x}_i, \boldsymbol{\theta}) \log |\boldsymbol{\Sigma}_k| \right\} \tag{44}$$

where $\gamma(\mathbf{x}_i, \boldsymbol{\theta})$ is a draw from $p(\gamma | \mathbf{x}_i, \boldsymbol{\theta})$. For both the partial predictive and PPN checks, we use $R = 200$ draws from the posterior predictive distributions.

Appendix B: Comparison to Bayes Factors

To compare two models, $\mathcal{M}_{\mathrm{A}}$ and $\mathcal{M}_{\mathrm{B}}$, the Bayes factor is

$$B_{AB} = \frac{p(\mathbf{X}|\mathcal{M}_{\mathrm{A}}) p(\mathcal{M}_{\mathrm{A}})}{p(\mathbf{X}|\mathcal{M}_{\mathrm{B}}) p(\mathcal{M}_{\mathrm{B}})}. \tag{45}$$

In the mixture model example, we approximate the marginal likelihood, $p(\mathbf{X}|\mathcal{M})$, with the harmonic mean of the likelihood values (Newton and Raftery, 1994):

$$\mathrm{p}(\mathbf{X}|\mathcal{M}) = \left\{ \frac{1}{R} \sum_{r=1}^{R} P(\mathbf{X}|\theta^{(r)}, \mathcal{M})^{-1} \right\}^{-1}, \tag{46}$$

where $\theta^{(r)} \sim p(\theta | \mathbf{X}, \mathcal{M})$ are draws from the posterior under model $\mathcal{M}$.

For the Gaussian mixture model example, the Bayes factors suggest $K = 3$, similarly to the PPN (Table 4).

For the multinomial mixture model example, the Bayes factors provide inconclusive evidence (Table 5).